Accessible Legal Assistance Agent

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Milestone 1 - Project Proposal

1. Motivation

This project develops an Accessible Legal Assistance Agent, a prototype of a retrieval-augmented, verifiable, and explainable generative AI system designed to help non-experts understand complex contractual and legal documents.

The project originates from the question implied by the phrase “Read the fine print.” In situations that involve significant legal consequences, such as signing contracts or service agreements, non-expert individuals often lack the ability to interpret complex legal language. Access to legal knowledge remains limited by both expertise and financial barriers. This work investigates how large language models (LLMs) can assist individuals in understanding such documents by retrieving relevant legal information and explaining difficult concepts without requiring costly or time-consuming consultations.

The proposed LLM-based agent specializes not only in retrieving relevant legal precedents and statutes but also in internally verifying the retrieved information, enabling users to access reliable and transparent legal references that support informed decision-making. For example, users concerned about the validity or fairness of contractual terms can consult relevant legal cases identified by the system. The model differentiates itself by operating as a ”tutor”, an educational legal assistant rather than a decision-maker. It translates complex terminology and explains legal reasoning in plain language, customized to the user’s specific context and concerns, especially for non-law-related individuals. Importantly, the agent is designed to avoid providing direct legal advice to prevent bias or unintended interpretations. Instead, it promotes independent, informed decision-making and perspective by guiding users toward relevant sources and transparent explanations.

2. Related Works

Barron et al. (2025) proposes an advanced Retrieval-Augmented Generation (RAG) framework that explicitly models the structural relationships among statutes, regulations, and case law. By incorporating legal ontologies into the retrieval pipeline, the system improves both the precision and contextual relevance of retrieved documents. This method effectively addresses a limitation of general-purpose RAG models, which often fail to capture the domain-specific dependencies present in legal texts. However, its primary focus remains on retrieval accuracy and knowledge grounding, and it does not provide clear explanations of how retrieved evidence supports the generated outputs. As a result, users, especially non-experts, may find it difficult to understand or verify the model’s reasoning.

Sadowski and Chudziak (2025) introduces a modular system in which specialized agents extract facts, formalize legal rules, and perform symbolic inference over structured knowledge graphs. The framework achieves high reasoning accuracy while maintaining transparency through verifiable intermediate representations. However, this method requires significant manual effort to design ontologies and may face scalability challenges when applied to diverse legal documents. Despite these limitations, it provides a valuable blueprint for constructing transparent, logic-driven legal reasoning systems.

Collenette et al. (2023) emphasizes transparency in computational legal reasoning. The authors introduce model-agnostic interpretability techniques that visualize reasoning chains, highlight influential textual evidence, and provide post-hoc justifications for AI-generated conclusions. These features substantially increase user trust and comprehension, fulfilling the ethical requirement of explainability in high-stakes domains. However, the system is primarily designed for predefined case datasets and lacks a dynamic retrieval component, limiting its scalability to diverse legal contexts.

3. Methodology

Integrating the strengths of the preceding approaches, the model will be developed as a single framework consisting of a four-stage pipeline. Raw contract documents are first segmented into clauses using tokenization and entity recognition, and each clause is embedded to support semantic retrieval. The retrieval module then employs a RAG pipeline to retrieve relevant statutes and case law, and the retrieved passages are reranked using a cross-encoder to ensure factual grounding. A multi-agent reasoning layer coordinates specialized agents (retriever, reasoner, and explainer) through a controlled reasoning chain. The Reasoner Agent constructs verifiable logical connections between the clause and retrieved evidence, while the Explainer Agent translates these reasoning steps into plain-language explanations. An explainability layer presents results as structured, plain language elaboration.

This methodology is appropriate because legal document interpretation demands both factual accuracy and logical transparency. The retrieval layer enables that all outputs are based on authoritative legal sources. The multi-agent layer introduces modular verification, where distinct agents cross-check retrieved evidence and reasoning consistency. The explainability layer translates this process into comprehensible, user-centered narratives.

Document retrieval and contextual augmentation are implemented using `LangChain`, while fine tuning of open source language models is performed with LoRA. To coordinate reasoning, the system employs the ReAct pattern together with lightweight multi-agent orchestration through `LangGraph`, which separates retrieval, reasoning, and explanation tasks for improved transparency. Prompt engineering techniques manage factual grounding and structural consistency, and reinforcement learning with human feedback (RLHF) may be applied to iteratively refine the clarity and quality of generated explanations.

Evaluation will assess both factual accuracy and interpretability. Retrieval performance will be measured by Precision and Recall, while reasoning validity will be evaluated through verification of intermediate reasoning chains to ensure that each generated statement is supported by retrieved evidence. Comparative baselines include a standard RAG model and a single-agent generation model, enabling quantitative assessment of the benefits introduced by verifiable multi-agent reasoning.

4. Dataset

The experimental corpus is derived from Contract Understanding Atticus Dataset (CUAD) (Hendrycks et al., 2021), a collection of legal contracts designed for clause-level comprehension and question answering. Each document in CUAD contains the full text of a contract segmented into clauses, accompanied by question–answer pairs that indicate the presence or absence of specific legal provisions. The dataset is provided in structured JSON format, where each record includes the contract text (`context`), a set of clause-level legal questions (`question`), annotated answer spans within the text (`answers`).

This dataset fits the proposed methodology because its clause-level annotations supervise retrieval and grounding for explainable reasoning. The question–answer format replicates real legal inquiries, allowing the retriever agent to reformulate queries and locate supporting evidence. Each annotated clause links directly to its contract text, enabling the reasoner agent to verify evidence and build logical chains and the explainer agent turn retrieved spans into clear, plain-language explanations.

Basic corpus-level analysis was performed to understand clause distribution and question diversity prior to model training. The average number of pages per contract was approximately 8–12 pages, with significant variation depending on contract type. (Figure 1) Each contract contains on average 40–50 clause-level annotations, corresponding to roughly 120 tokens per clause. (Figure 2) A total of 41 clause categories were annotated, such as Confidentiality, Indemnification, Termination, and Governing Law. (Figure 3) The dataset includes 23 distinct contract types (Service Agreement, Lease Agreement, Stock Purchase Agreement) displays the number of contracts per contract category. (Figure 4)

5. Work Plan

Contributions

Hyemin Kim	Researched relevant sources (Related Works)
Hyunkyu Park	Proposed ideas and approaches (Motivation and Methodology)
Zalopher Zhang	Conducted dataset research and EDA (Dataset)

Weekly Plan

Date	Planned Tasks and Deliverables
Oct 12, 2025	Milestone 1 Submission (Everyone)
Oct 19, 2025	Revise proposal based on feedback (Hyunkyu)
	Finalize dataset selection and preprocessing plan (Hyemin, Zalopher)
Oct 26, 2025	Implement RAG retrieval module (Everyone)
Nov 02, 2025	Develop multi-agent reasoning chain using ReAct (Everyone)
Nov 09, 2025	Evaluate and revise: retrieval and reasoning layer (Everyone)
Nov 16, 2025	Milestone 2 Submission (Everyone)
Nov 23, 2025	Fine-tune explanation model with LoRA (Zalopher)
	Integrate explainability layer (Hyemin, Hyunkyu)
Nov 30, 2025	Evaluate and revise: reasoning verification and explainability (Everyone)
Dec 07, 2025	Prepare final documentation (Everyone)
Dec 14, 2025	Milestone 3 Submission (Everyone)

References

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Joe Collenette, Katie Atkinson, and Trevor Bench-Capon. Explainable ai tools for legal reasoning about cases: A study on the european court of human rights. *Artificial Intelligence*, 317:103861, 2023. ISSN 0004-3702. doi: https://doi.org/10.1016/j.artint.2023.103861. URL https://www.sciencedirect.com/science/article/pii/S0004370223000073.

Dan Hendrycks, Collin Burns, Anya Chen, and Spencer Ball. Cuad: An expert-annotated nlp dataset for legal contract review. *NeurIPS*, 2021.

Albert Sadowski and Jarosław A. Chudziak. On verifiable legal reasoning: A multi-agent framework with formalized knowledge representations, 2025. URL https://arxiv.org/abs/2509.00710.

Appendix A. Figures

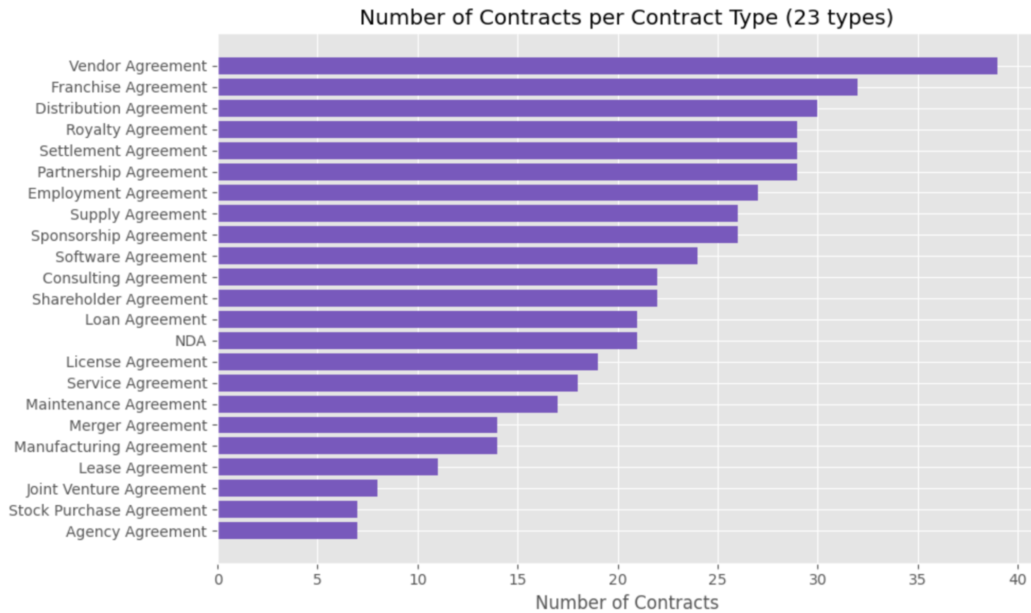

Figure 1: Number of Contracts per Contract Type

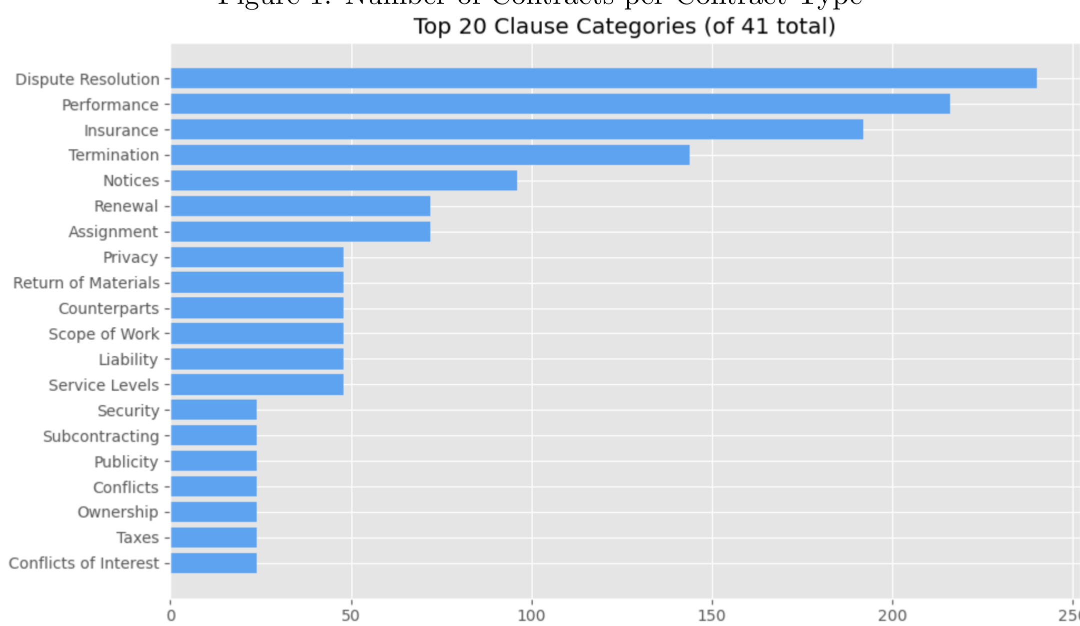

Figure 2: Top 20 Clause Categories

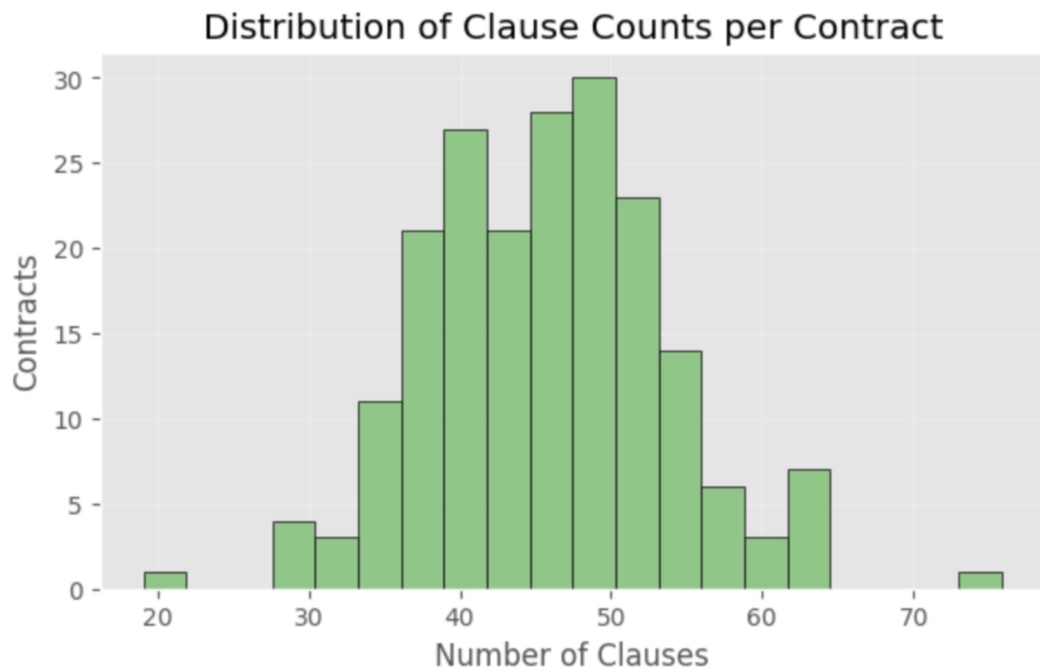

Figure 3: Distribution of Clause Counts per Contract

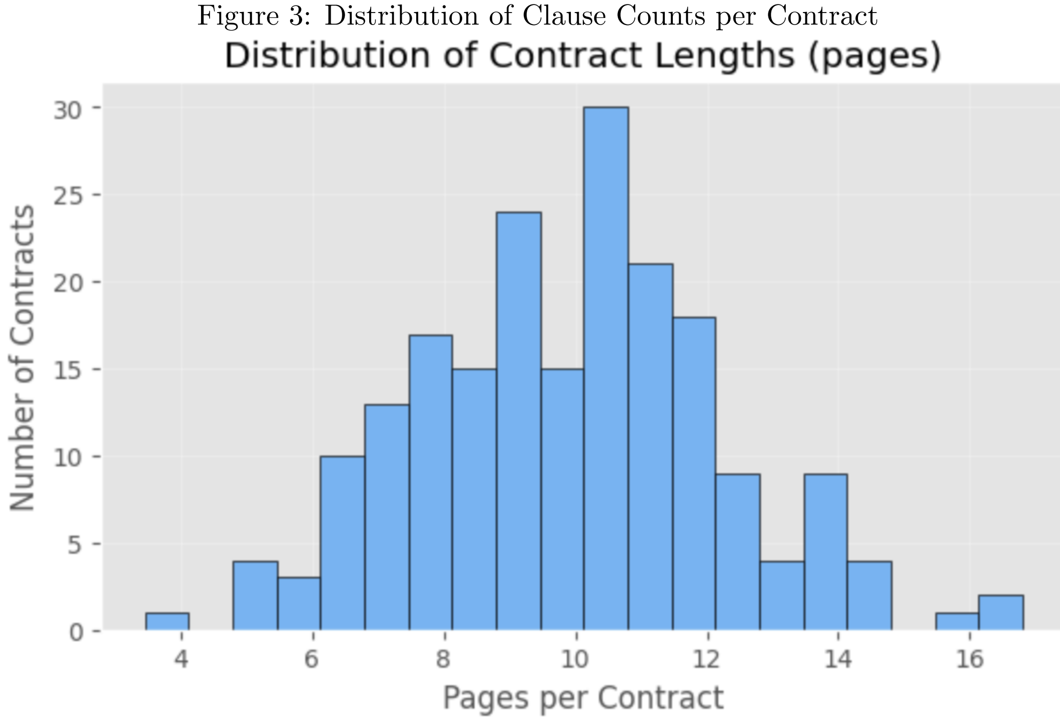

Figure 4: Distribution of Contract Lengths (pages)